

FilingAlpha — SEC Filing Signal Research Note

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Abstract

A backtest of classical natural-language signals extracted from SEC filings, evaluated against market-adjusted forward returns with strict filing-lag (no lookahead). Universe: 60 companies; filings: 400 10-K, 1182 10-Q; sample span: 2015-10-23 to 2026-07-02. The strongest 10-K signal is negative tone (Loughran-McDonald) at a 21-day horizon (IC -0.447, long-short spread -6.08% net of cost, $t = -2.66$).

1 Methodology

Each signal is computed per filing and aligned to the firm's forward return over 21- and 63-trading-day horizons, entered on the first trading day strictly after the filing date. Returns are market-adjusted (excess of SPY over the identical window). Signals are evaluated by Spearman information coefficient (with a t -statistic) and by an event-study long-short tercile spread, net of 10 bps per side. 10-K and 10-Q filings are evaluated separately, since the text-change signals are an annual-report phenomenon. A significance marker of * denotes $|t| \geq 1.5$ and ** denotes $|t| \geq 2$.

2 Results

Form	Signal	Horizon	IC	IC t	L-S %	Spread t
10-K	negative tone (Loughran-McDonald)	21d	-0.447	-3.38	-6.08	-2.66**
10-K	Gunning-Fog readability	21d	-0.187	-0.93	-4.06	-2.24**
10-K	negative tone (Loughran-McDonald)	63d	-0.090	-0.37	-6.15	-1.55*
10-K	litigious tone	21d	-0.004	-0.02	-2.40	-1.20
10-K	uncertainty tone	21d	-0.059	-0.25	-1.92	-0.82
10-K	year-over-year filing similarity (Lazy Prices)	63d	-0.051	-0.90	-2.49	-0.51
10-K	year-over-year filing similarity (Lazy Prices)	21d	-0.113	-2.02	-1.39	-0.48
10-K	litigious tone	63d	-0.013	-0.08	-1.34	-0.34
10-K	uncertainty tone	63d	0.204	0.67	-0.97	-0.20
10-K	risk-factor section change	63d	0.010	0.17	-0.60	-0.11
10-K	risk-factor section change	21d	0.064	1.12	-0.31	-0.05
10-K	Gunning-Fog readability	63d	-0.203	-0.99	-0.17	0.01
10-Q	Gunning-Fog readability	63d	0.089	1.43	7.20	3.08**
10-Q	negative tone (Loughran-McDonald)	63d	-0.018	-0.28	6.27	2.83**
10-Q	negative tone (Loughran-McDonald)	21d	0.020	0.31	2.04	2.25**
10-Q	Gunning-Fog readability	21d	0.014	0.21	2.32	2.21**
10-Q	uncertainty tone	63d	0.008	0.11	5.22	2.17**
10-Q	year-over-year filing similarity (Lazy Prices)	63d	-0.064	-1.02	3.19	1.33
10-Q	uncertainty tone	21d	0.031	0.44	1.01	1.01
10-Q	year-over-year filing similarity (Lazy Prices)	21d	-0.068	-1.13	0.50	0.59
10-Q	litigious tone	21d	-0.015	-0.21	0.00	0.16
10-Q	litigious tone	63d	-0.113	-1.93	0.11	0.12

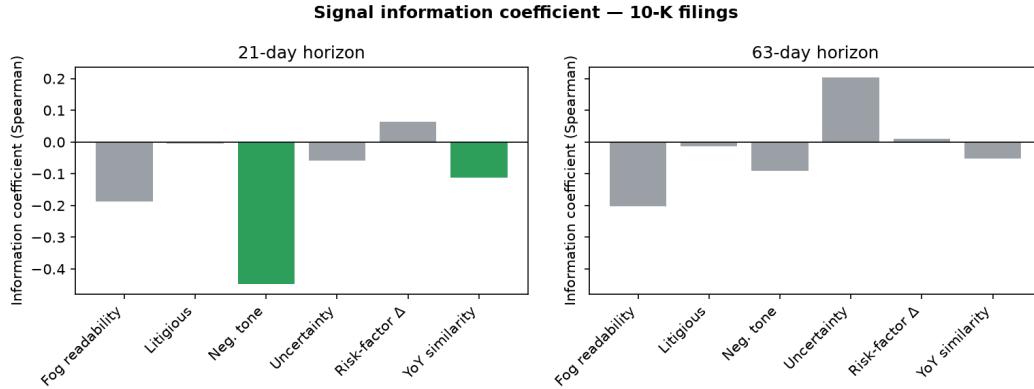


Figure 1: Information coefficient by signal (10-K). Bars coloured by t-statistic magnitude: green significant, amber suggestive, grey noise.

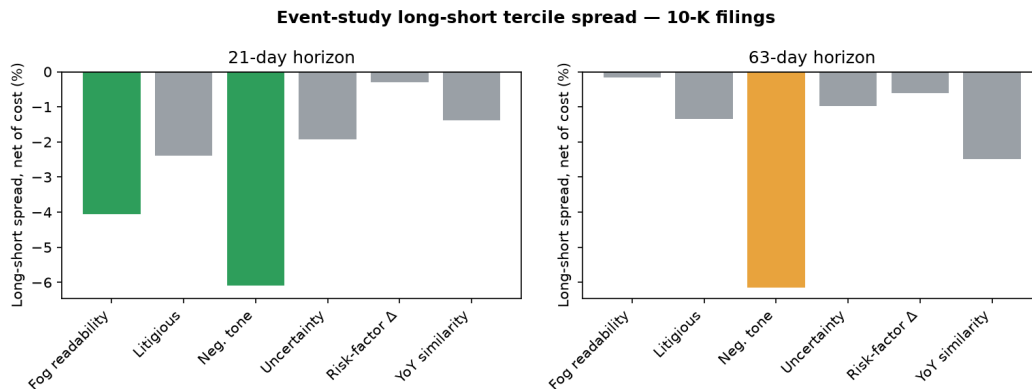


Figure 2: Event-study long-short tercile spread by signal (10-K), net of cost.

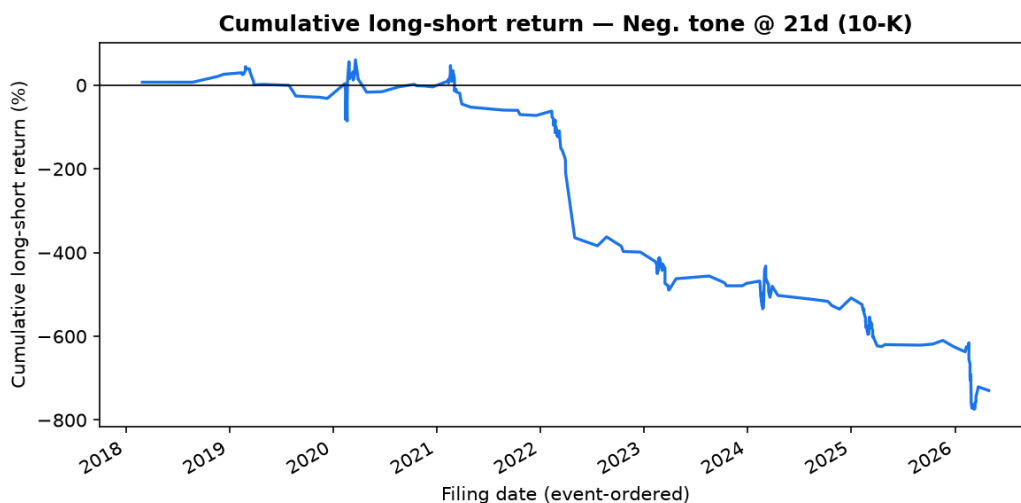


Figure 3: Event-ordered cumulative long-short return for the headline signal. A visualisation of the tercile spread, not a tradeable curve.

3 Interpretation

Across a universe of 60 companies (400 10-K, 1182 10-Q), each signal was evaluated against market-adjusted forward returns, with 10-K and 10-Q filings measured separately. Returns are excess of SPY over an identical holding window, and the long-short construction is charged 10 bps per side.

On 10-K filings, the strongest result is negative tone (Loughran-McDonald) at a 21-day horizon: an information coefficient of -0.447 ($t = -3.38$) and a long-short tercile spread of -6.08% net of cost ($t = -2.66$) over 374 filing events. The sign and magnitude are economically sensible, but the result is statistically significant at the 95% level — consistent with a real but small anomaly that this universe is too narrow to resolve.

The expanding-window walk-forward classifier confirms this: out-of-sample accuracy of 0.541 and AUC of 0.514 over 255 held-out filings is indistinguishable from a coin flip. The contribution here is the leakage-free measurement harness — an honest null is the correct, credible outcome at this scale, not a failure.

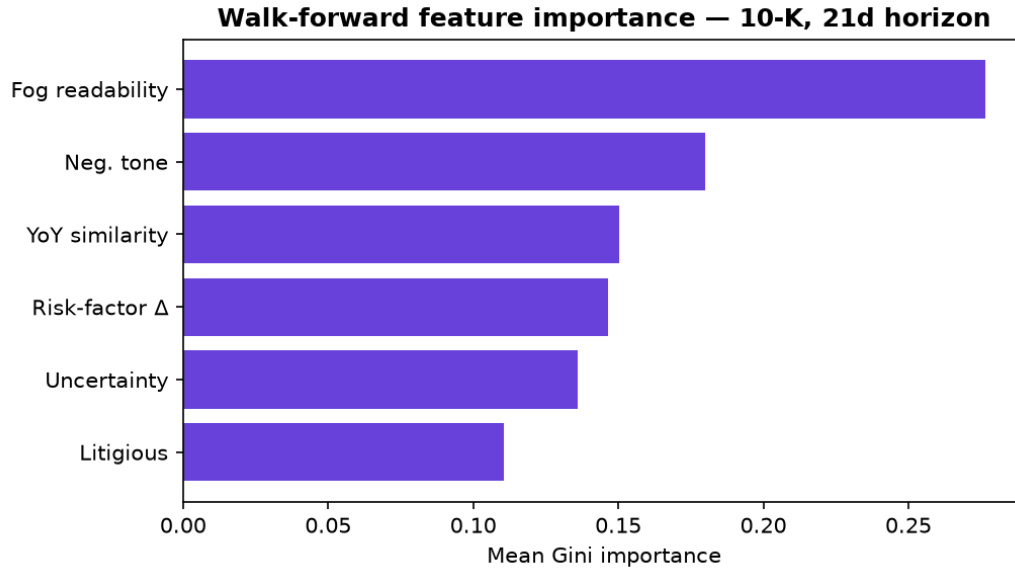


Figure 4: Walk-forward model feature importances (10-K).

4 Honest caveats

This is a measurement harness, not a deployed strategy. At this universe size the results are not expected to reach statistical significance; the value is the leakage-free, point-in-time-correct construction and an honest reporting of weak or null effects. Scaling the universe to several hundred small- and mid-cap firms is the principled path to significance and is left as future work.